

# Malek Badawi

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## EDUCATION

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**Louisiana State University (LSU), Baton Rouge, LA**

**Expected May 2028**

*Bachelor of Science in Computer Science (Software Engineering)*

*GPA: 4.17/4.30*

**Minors:** Mathematics and Honors Research

**Honors:** Stamps Scholar (full-ride national scholarship), President's Future Leaders in Research, President's Honor Roll

**Coursework:** LLM Application Development (Fall 2026) | Advanced Data Structures and Algorithms, Discrete Structures, Numerical Methods, Computer Organization and Design, Differential Equations/Linear Algebra, Object-Oriented Programming

**University College London** – Artificial Intelligence Concepts & Applications, Summer 2025 (Distinction)

## EXPERIENCE

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**Undergraduate Researcher**

**March 2026 – Present**

*Pennington Biomedical Research Center – Imaging Department*

*Baton Rouge, LA*

- Cross-checked the lab's BRISQUE QC pipeline against manual review and showed it ranked motion-corrupted scans among the highest, due to natural-image statistics missing MRI artifacts, and drove the pivot to radiologist-validated MRIQC.
- Consolidated FreeSurfer cortical-surface outputs for approximately 460 subjects into a unified symlink-based directory, automating version-conflict resolution across inconsistent naming conventions and eliminating case-by-case file selection.
- Deployed MRIQC in an Apptainer container on a shared Linux/NFS server and built a BIDS-compliant pilot dataset for 23 subjects, including DICOM-to-NIfTI conversion, symlink organization and programmatic validation with pybids.
- Developed Bash and MATLAB batch-processing workflows across seven concurrent studies containing 500+ subjects, adding dry-run validation and failure safeguards while resolving file-processing and variable-scope errors.

**Software & Data Engineering Intern**

**September 2025 – June 2026**

*Automotive Data Analytics*

*Remote*

- Built and tested a three-endpoint SMS authentication service using JavaScript, Azure Logic Apps, Azure Table Storage and Twilio, supporting OTP requests, verification and session validation with five-minute code expiration and 24-hour sessions.
- Designed isolated Logic App workflows for each authentication endpoint and restructured token storage to enforce one active session per user after identifying that expired tokens were not being removed from Azure Table Storage.
- Built Microsoft Fabric ETL pipelines that consolidated 10,000+ customer records from multiple dealership systems into a centralized lakehouse, automated recurring refreshes and supported Power BI reports that replaced manual performance-report compilation.

## PROJECTS

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**Golf Swing Analyzer** | *Python, OpenCV, MediaPipe*

- Developed a Python computer-vision pipeline using OpenCV and MediaPipe to extract body landmarks from swing videos and calculate hip rotation, shoulder rotation, spine angle and tempo on a frame-by-frame basis.

**Personal Portfolio Site** | *React, Vite, React Router, CSS, Vercel*

- Designed and built a single-page application portfolio with client-side routing and a custom design system; deployed with CI on push at [malekbadawi.com](http://malekbadawi.com).

## SKILLS

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**Languages:** Java, Python, JavaScript, C, SQL, Bash, HTML/CSS, MATLAB

**Frameworks & Libraries:** React, Node.js, OpenCV, MediaPipe, NumPy, pandas

**Tools & Platforms:** Git/GitHub, Linux/Unix, REST APIs, Postman, Azure (Logic Apps, Table Storage), Microsoft Fabric, Power BI, Twilio, Vercel, FreeSurfer, VS Code, Apptainer/Singularity, pybids, BIDS, NIfTI, DICOM

**Spoken:** Arabic (fluent), English (fluent), Spanish (conversational)

## LEADERSHIP

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**Student Government Senator**

**August 2025 – Present**

*Louisiana State University – Baton Rouge, LA*

- Elected by the College of Engineering; serve on the Student Auxiliary Services and Academic Affairs Committees, contributing to initiatives involving campus technology resources and ethical AI policy.